XRP Engineering Design Process

Facilitator Guide

Module Information

Module Overview

MODULE LENGTH | 460 MINUTES (ABOUT 8 HOURS)

MODULE DESCRIPTION

Participants in the Engineering Design Process module will learn the engineering design process through hands-on activities, including brainstorming, sketching, prototyping, and decision-making. They will design and refine solutions, such as a gripper mechanism, using iterative testing and research. In the final challenge, students will create a robot to solve a custom game's challenges and present their designs, practicing feedback incorporation to improve their work.

MODULE PREREQUISITES

- Explore the XRP Getting Started Resources for building and coding
- XRP Python Programming the Drivetrain or XRP Blocks Programming the Drivetrain

SETUP

For every two students you have, you will need one XRP robot and its necessary supplies for operation (cables, batteries, etc.). The XRP robots will need to be already assembled, and students will need some prior knowledge about them. See the Module Prerequisites above for more information.

The facilitator will need to have at least one 3D printer with access to PLA filament for printing all the materials needed prior to the start of the module. Use the 3D Printing Guide for more information. Each lesson has its own specific setup instructions provided at the beginning of the content.

Here is an Evidence Plan / Standards Alignment for the module.

MATERIALS

The following is the list of all the materials you will need to complete this module that are not digitally provided to you. Each lesson has its own specific material list provided at the beginning of the content.

- Drinking straws (non-bendy)
- Plastic spoons
- Cardboard sheets/scraps
- Popsicle sticks
- Cardstock
- 8.5" x 11" paper
- Rubber bands
- Masking tape
- Hot glue
- Bean bag
- Toy cars
- 1 kg mass
- Fishing line/string
- Scissors
- Electronic device with access to the internet
- Paper
- Pens/pencils/markers
- 3D-Printed Primitives Set
- 3D-Printed Bar Set
- Small objects to pick up (soda can, cube, ball, etc.)
- XRP Robot (one robot per two students)
- AA batteries (four batteries per XRP Robot)
- Ping-pong balls
- Plastic cups
- Poster board

DOWNLOADABLE RESOURCES

Below is the list of all the downloadable resources you will need to complete this module that are not digitally provided to you. Each lesson has its own specific downloadable resources list provided at the beginning of the content.

• Evidence Plan/Standards Alignment
• Student Slide Decks
  ○ Lesson 1
  ○ Lesson 2
  ○ Lesson 3
  ○ Lesson 4
  ○ Lesson 5
  ○ Lesson 6
  ○ Lesson 7
  ○ Lesson 8
• Engineering Design Process Graphic
• See-Think-Wonder Template
• Engineering Design Process Flashcards
• Decision Matrix Template
	• Engineering Design Grippers
	• 3D Printing Guide
	• Printed Engineering Design Primitives Shapes
	• Printed Engineering Design Grippers
	• Picking Up Pixels video
	• Stationary Gripper Engineering Drawing
	• Geared Angular Gripper Engineering Drawing
	• Radial Gripper Engineering Drawing
	• Mechanisms Flashcards
	• Java The Hutts Robot Reveal Centerstage video
	• Resource for Creating Mind Maps Digitally
	• Rapid Prototyping video
	• Engineering Drawing video
	• Tips to Improve Your Presenting Skills video

ENGINEERING DESIGN PROCESS

FIRST teams engage in hands-on building, programming, and problem-solving, applying engineering and physics principles to design and improve functional robots. Throughout the season, teams set goals, manage their project timeline, and reflect on their performance to refine their robots and strategies. Collaboration and communication are key, with team members assuming various roles like builders, programmers, and designers. Teams present their work to judges and the public, sharing their design choices and strategies. The competition encourages continuous improvement through iterative testing and problem-solving, with teams overcoming challenges both on and off the field to optimize performance and reliability.

Lesson Outline

LESSON 1 | ENGINEERING DESIGN PROCESS INTRODUCTION

Assess your problem-solving and initial understanding of the engineering design process and how to guide others through this process.

LESSON 2 | BRAINSTORMING STRATEGIES AND IDEATION

Review the specific strategies for brainstorming such as the Systemic Approach to Ideation, Mind Mapping, and general methods of brainstorming.

LESSON 3 | SKETCHING AND DRAWING

Properly draw and sketch primitive 3D shapes in an isometric view.

LESSON 4 | RESEARCHING KNOWN SOLUTIONS

Observe a professional model based on an existing solution to a problem, such as a robot design from a prior *FIRST* robot.

LESSON 5 | GRIPPER DECISIONS

Utilize a decision matrix and design a claw mechanism to grab a small item.

LESSON 6 | PROTOTYPING AND ITERATION

Delve into the iterative process of rapid prototyping emphasizing the importance of experimentation and learning from failures.

LESSON 7 | CULMINATING CHALLENGE

Utilize the engineering design process to develop a game and design a robot to solve its challenges.

LESSON 8 | PRESENT AND GET FEEDBACK

Present your final design and learn about feedback techniques.

Lesson 1 | Engineering Design Process Introduction

Introduction

OVERVIEW

In this lesson, you will assess students' problem-solving and initial understanding of the engineering design process and how to guide them through the process. Students will complete an engineering design challenge while learning important collaboration and communication methods for facilitating this with others.

LEARNING OUTCOMES

In this lesson, learners will:

- Understand how to actively engage in observations, share thoughts, and ask questions to better understand a task.
- Understand how to collaborate effectively with others by setting clear goals, working toward them, and assessing our progress and teamwork.
- Understand how to present as a group and reflect, using feedback to understand the steps of the engineering design process and how our work aligns with it.

Prepare

MATERIALS

Each team will need:

- Engineering Design Process graphic
- Engineering Design Process flashcards
- See-Think-Wonder template
- Pens/pencils/markers
- Based on the design challenge chosen:
    - Drinking straws (non-bendy)
    - Plastic spoons
    - Cardboard sheets/scraps
    - Popsicle sticks
    - Cardstock
    - 8.5" x 11" paper
    - Rubber bands
    - Masking tape
    - Hot glue
    - Bean bag
    - Toy cars
    - 1 kg mass
    - Fishing line/string
    - Scissors

DOWNLOADABLE RESOURCES

- Student Facing Slide Deck
- Engineering Design Process Graphic
- See-Think-Wonder Template
- Engineering Design Process Flashcards

SETUP

- Select which of the three design challenges you will use in this lesson. Gather all the necessary materials to complete that activity. These materials can be found in the notes of the student slide deck for each activity.
- Print or digitally share the downloadable resources with the students.
- Have the student-facing slide deck ready to present.

- Make sure you have a method for students to document and share their design challenge.

Learn and Apply

ENGINEERING DESIGN PROCESS (5 MINUTES)

Review the Engineering Design Process graphic with the students, discussing each step.

An alternative teaching method could be to divide the students into small groups, assign them a portion of the process, and have them report out. Students can share a real-world example of when they have done this part of the process.

INQUIRY ACTIVITY

Pass out the See-Think-Wonder template to each student. Discuss the three columns of the document.

Have a variety of materials set out on the table. Ask students to fill in the document based on what materials they see in front of them and their new knowledge of the engineering design process.

You can choose to have the students do this on their own or in small groups.

CLASSROOM IMPLEMENTATION

See-Think-Wonder

This technique is used to engage students in the inquiry process. Its purpose is to generate questions and ideas about what they might already know and what they might learn.

Forming Groups

Use intentional grouping methods when working with students to create groups for a task. A few methods include:

- Interest-based grouping: Groups students according to passion and interest to increase engagement. You can use tasks like “4 corners” to place them into groups.
- Random grouping: Use tools like random number generators, card drawers, or apps to assign groups. This promotes social interaction and prevents cliques, which promotes inclusion in *FIRST*.
- Task-based grouping: Group students according to the tasks or roles needed for a specific project or activity. Ensures that all roles and tasks are covered and teaches students about teamwork and collaboration.

GOAL SETTING (5 MINUTES)

The SMART method was developed to improve corporate planning and goal setting by setting clearer and more attainable targets. Using SMART goals can help you focus your efforts and increase your chances of achieving your goal.

Review the parts of a SMART goal with the students and have them write their own SMART goal for the design challenge.

DESIGN CHALLENGE (30 MINUTES)

Based on the materials you have available, review one of the three design challenge options for the students will work on.

If you have not already, group up the students to work on the design challenge. Have the students work in groups to complete the design challenge. Remember that they will be on a time constraint; it does not matter if they are successful in completing the activity but rather that they are using the engineering design process.

ITERATION (5 MINUTES)

After students have had some time to work through their engineering challenge, have them create an additional SMART goal around improving their solution.

PRESENTATION | SHARE OUT (10 MINUTES)

Students will present their end result in the stage of completion they are at in the time constraint. This can be just verbal or include visuals and documentation. They may have a working model or a failing one.

CLASSROOM IMPLEMENTATION

You can print the Engineering Design Process flashcards and give them to students to help them guide the process. The questions are also here for your reference as you are guiding students.

Define the Problem

- What problem are you trying to solve?
- Why is this problem important to address?
- Who will benefit from solving this problem?
- What are the constraints and criteria for your solution?

Research and Gather Information

- What do you already know about this problem?
- What additional information do you need?
- Where can you find reliable information about this topic?
- What existing solutions or approaches have been tried?

Brainstorm Possible Solutions

- What are some potential ways to solve this problem?
- How can you think outside the box to produce creative solutions?
- What are the pros and cons of each potential solution?
- How can you combine or modify ideas to improve them?

Select the Best Solution

- Which solution do you think is the best? Why?
- How does this solution meet the criteria and constraints?
- What are the potential challenges or risks with this solution?
- How will you know if this solution is successful?

Develop and Prototype

- What materials and tools will you need to build your prototype?
- What steps will you take to create your prototype?
- How will you ensure accuracy and precision in your prototype?
- How will you document the process of building your prototype?

Test and Evaluate

- How will you test your prototype to see if it works?
- What criteria will you use to evaluate your prototype?
- What data will you collect during testing?
- How will you analyze the results of your tests?

Improve and Refine

- What did you learn from testing your prototype?
- What worked well and what didn't?
- How can you improve your solution based on the test results?
- What changes will you make to your prototype?

Communicate Results

- How will you present your solution and findings?
- What evidence will you use to support your conclusions?
- How will you explain the process you followed and the decisions you made?
- Who is your audience and how can you tailor your presentation to them?

General Reflection Questions

- What was the most challenging part of the process for you?
- What did you learn about teamwork and collaboration?
- How did you overcome any obstacles or setbacks?
- What skills did you develop or improve during this project?
- How could you apply what you have learned to future projects or problems?

Evaluate and Reflect

EVALUATE (5 MINUTES)

Options to evaluate students' learning can include:

- Knowledge check questions found in the LMS course
- Presentation of design challenge
- Responses, written or verbal, to the Engineering Design Process flashcard questions
- Response to the following questions
    - What might you need to know to fully understand the engineering design process?
    - What might you need to do to help guide others through understanding their accomplishments and what to do next?
    - How is iteration important? What is needed to guide continued iteration??

REFLECT

Utilize the Fist to Five method, or another method of your choosing, for a quick way to gauge your student's confidence with this step of the engineering design process.

Lesson 2 | Brainstorming Strategies and Ideation

Introduction

OVERVIEW

In this lesson, you will dive into the specific strategies for brainstorming such as the systemic approach to ideation, mind mapping, and other general methods of brainstorming.

LEARNING OUTCOMES

In this lesson, learners will:

- Understand how to use structured brainstorming methods to identify and describe various problems and potential solutions without filtering ideas.
- Create visual representations of ideas using mind-mapping techniques to organize and connect concepts effectively and guide others in the process.
- Guide others in brainstorming sessions to generate, evaluate, and select the best ideas using structured methods.

Prepare

MATERIALS

Each team will need:

- Electronic device with access to the internet
- Paper
- Pencils/pens/markers

DOWNLOADABLE RESOURCES

- Student Facing Slide Deck
- Resource for Creating Mind Maps Digitally

SETUP

- Make sure you have a method for students to document and share their mind map.
- Print or digitally share the downloadable resources with the students.
- Have the student-facing slide deck ready to present.

Learn and Apply

INQUIRY ACTIVITY (10 MINUTES)

As students enter the classroom, challenge them to brainstorm a list of their favorite foods (or any other criteria that you decide).

Have the students get into small groups and share their lists. Have them organize their lists into different categories such as dessert, drinks, main dish, etc.

EFFECTIVE BRAINSTORMING (5 MINUTES)

The purpose of brainstorming is to generate as many ideas as possible without limits. Filtering and analysis of ideas can happen, but not until the original generation of ideas occurs.

Review the general tips for brainstorming with students.

SYSTEMIC IDEATION (5 MINUTES)

Walk through the systemic approach to ideation with students.

Consider the following questions that might help you guide students in this process:

- How is brainstorming through systems different from a more open brainstorming method?
- How might people struggle with this process if they are not used to brainstorming this way?
- How can thinking about concepts within systems benefit the brainstorming process?

SYSTEMIC IDEATION CHALLENGE (10 MINUTES)
Using one of the following prompts, or creating one of your own, have the students work together in small groups to work through the systemic ideation brainstorming process. Possible prompts include:

- Building a factory to produce a product
- Starting a business to sell a product internationally
- Creating a product that can solve a local problem

CLASSROOM IMPLEMENTATION
Supplies: Students will need supplies to document their systemic ideation process on paper or digitally.

Grouping: Use one of the following methods or your own to break students into groups for the challenges.

- Random grouping based on count-off
- Intentional grouping based on knowing students
- Predetermined grouping methods, such as table groups

MIND MAPPING (5 MINUTES)
Talk through the process of developing a mind map, and utilize a topic you are familiar with to use as an example with students.

CHALLENGE | APPLICATION (15 MINUTES)
Students will now work, either in groups or alone, to develop their own mind map on a topic of your choosing. After a set time, students will present their mind maps to the class. Students can give feedback to each other by asking clarifying questions, sharing values/appreciations for solutions, and offering concerns or suggestions about the problems and solutions.

CLASSROOM IMPLEMENTATION
Consider the following questions that might help you guide students:

- What observations did you make about your group when brainstorming?
- How would you rate the quality of your group's brainstorming?
- What other strategies did your team use when brainstorming?

Evaluate and Reflect

EVALUATE (5 MINUTES)
Options to evaluate students' learning can include:

- Knowledge check questions found in the LMS course
- Presentation of their mind map
- Response to the questions under classroom implementation section above
- Response to the following questions
  - Why is reflection important?
  - How can framing reflection questions provoke deeper thought and improve learning?
  - How is systemic brainstorming different from other types of brainstorming?

REFLECT
Utilize the Fist to Five method, or another method of your choosing, for a quick way to gauge your student's confidence with this step of the engineering design process.

Lesson 3 | Sketching and Drawing

Introduction

OVERVIEW

In this lesson, you will learn how to properly draw and sketch primitive 3D shapes such as cubes, cylinders, and prisms isometrically.

LEARNING OUTCOMES

In this lesson, learners will:

- Develop strategies for identifying and outlining the key elements of a concept sketch using basic shapes and gradually adding details to create a refined concept drawing.
- Develop strategies for using shading techniques such as hatching, cross-hatching, and stippling to create depth and dimension in sketches and accurately represent light sources and shadows.
- Develop strategies for collaborating with peers to interpret and recreate 3D structures from sketches, providing and receiving constructive feedback to improve clarity and detail in drawings.

Prepare

MATERIALS

Each team will need:

- 3D-Printed Primitives Set (*these will need to be 3D printed in advance, one set per 2 students)*
- Graph paper
- Paper/pencils/markers

DOWNLOADABLE RESOURCES

- Student Facing Slide Deck
- 3D-Printed Primitives Set
- 3D Printing Guide

SETUP

- Have the 3D-Printed Primitives Set printed and divided out into sets *(one set per two students)*.
- Print or digitally share the downloadable resources with the students.
- Have the student-facing slide deck ready to present.
- Make sure you have a method for students to document and share their concept sketches.

Learn and Apply

INQUIRY ACTIVITY (10 MINUTES)

As students enter the classroom, instruct them to take the first 10 minutes to select a 3D-printed primitive or two and sketch it on some graph paper as detailed as possible. Have students share with another and get feedback. Can they tell which primitive you drew? What could you improve on?

ELEMENTS OF CONCEPT SKETCHING (5 MINUTES)

Review the steps to concept sketching, showing examples of each of the different elements.

CREATING A STRUCTURE CHALLENGE (30 MINUTES)

Students will select three 3D primitives and build a structure then create a concept sketch, iterating until they think they have a clear sketch. Have the students provide their sketch to another student and see if they can build their structure based only on the concept sketch.

CLASSROOM IMPLEMENTATION

Here are a few feedback strategies you can teach your students to use with one another:

Ladder of Feedback

- Clarify: Presenters ask specific questions or clarifications.
- Value: The peer reviewer starts with what they value or like about the work.
- Concerns: The peer reviewer then mentions any concerns or areas that need improvement.
- Suggestions: Finally, the peer reviewer offers suggestions for enhancement.

Two Stars and a Wish

- Description: A straightforward method where feedback is given in a balanced way.
- Steps:
    1. Two Stars: The peer reviewer identifies two aspects they liked or found impressive.
    2. A Wish: The peer reviewer shares one area they wish to see improved or developed further.

Glow and Grow

- Description: A simple feedback strategy focusing on strengths and areas for improvement.
- Steps:
    1. Glow: The peer reviewer highlights something that "glows" (stands out positively) in the work.
    2. Grow: The peer reviewer points out an area where the student can "grow" (improve or develop).

Question-Comment -Suggestion

- Description: A concise feedback approach where peers engage with the work through questioning, commenting, and suggesting.
- Steps:
    1. Question: The peer reviewer asks one question about the work to gain deeper understanding or clarification.
    2. Comment: The peer reviewer makes one comment about what they found interesting or noteworthy.
    3. Suggestion: The peer reviewer provides one suggestion for improvement or further exploration.

Evaluate and Reflect

EVALUATE (5 MINUTES)

Options to evaluate students' learning can include:

- Knowledge check questions found in the LMS course
- Presentation of their concept sketches
- Response to the following questions
    - What details help others understand a sketch?
    - Which methods or techniques did you find more challenging and why?
    - What are some key takeaways after practicing sketching?

REFLECT

Utilize the Fist to Five method, or another method of your choosing, for a quick way to gauge your student's confidence with this step of the Engineering Design Process.

Lesson 4 | Researching Known Solutions

Introduction

OVERVIEW

In this lesson, you will observe a professional model based on an existing solution to a problem, such as a robot design from a prior *FIRST* robot.

LEARNING OUTCOMES

In this lesson, learners will:

- Learn the basic mechanism found in *FIRST* Tech Challenge and *FIRST* Robotics Competition robots.
- Observe and take notes on a professional robot model to better understand engineering design.
- Research and present information about a robot part, including how it is used and its pros and cons.
- Use online research skills to find good information about robot parts and create a clear presentation of findings.

Prepare

MATERIALS

Each team will need:

- Pens/pencils/markers
- Paper
- Computer with internet access

DOWNLOADABLE RESOURCES

- Student Facing Slide Deck
- Mechanisms Flashcards
- Java the Hutts Robot Reveal Centerstage
- See Think Wonder Template

SETUP

- Print or digitally share the downloadable resources with the students.
- Have the student-facing slide deck ready to present.
- Make sure you have a method for students to document and share their mechanism research.

Learn and Apply

MECHANISMS OF FIRST ROBOTS (10 MINUTES)

Provide the students with the See-Think-Wonder template as you walk through the six types of mechanisms (either using the flashcards or the student slide deck) and have the students record their thinking.

Show the “Java the Hutts Robot Reveal Centerstage” video and have the students record their thinking in the template as well. Have students share their thoughts with one another.

RESEARCH TECHNIQUES (5 MINUTES)

Review the tips for effective Google searching, showing examples for students using your internet browser.

RESEARCH CHALLENGE (30 MINUTES)

Break the students into small groups, assign each group a *FIRST* robot mechanism. Provide them with time to research their mechanism and put together a presentation that covers the following:

- Describe the mechanism and common uses in *FIRST*.
- Find various examples of the mechanism in the real world.
- List key features, advantages, and challenges of the mechanism.
- Sketch and example of how you could use this mechanism.

Groups will then share their findings with the class.

Evaluate and Reflect

EVALUATE (5 MINUTES)

Options to evaluate students' learning can include:

- Knowledge check questions found in the LMS course
- Presentation of their mechanism research
- Response to the following questions
    - What were the most useful sources?
    - What challenges did you encounter? How did you overcome them?
    - Were there mechanisms you did not recognize? What specifically did you learn about those mechanisms?

REFLECT

Utilize the Fist to Five method, or another method of your choosing, for a quick way to gauge your student's confidence with this step of the engineering design process.

Lesson 5 | Gripper Decisions

Introduction

OVERVIEW

In this lesson, you will utilize a decision matrix and design a claw mechanism to grab an item like a bottle.

LEARNING OUTCOMES

In this lesson, learners will:

- Use a decision matrix to evaluate different options and make informed decisions.
- Work with a group to design and create a claw mechanism, using a decision matrix to choose the best design.
- Present a group design, receive feedback from others, and reflect on the effectiveness of the chosen design.

Prepare

MATERIALS

Each team will need:

- 3D-Printed Bar Set (one set per two students)
- Paper
- Pens/pencils/markers
- Rubber bands
- Masking tape
- Small objects to pick up (soda can, cube, ball, etc.)

DOWNLOADABLE RESOURCES

- Student Facing Slide Deck
- See Think Wonder Template
- 3D Printing Guide
- Grippers
- Picking Up Pixels video
- Stationary Gripper Engineering Drawing
- Geared Angular Gripper Engineering Drawing
- Radial Gripper Engineering Drawing
- Decision Matrix Template

SETUP

- Have the 3D Printed Bar Set printed and divided out into sets *(one set per 2 students)* – using the 3D Printing Guide.
- Print or digitally share the downloadable resources with the students.
- Have the student-facing slide deck ready to present.
- Make sure you have a method for students to document and share their decision matrix.

Learn and Apply

GRIPPER DESIGNS (10 MINUTES)

Hand out the See-Think-Wonder template to students. Have them fill out the template while they watch the Picking Up Pixels video. Utilize the guiding questions in the Classroom Implementation section below. At the conclusion of the video, students share their findings with others.

DEVELOP A GRIPPER CHALLENGE (30 MINUTES)

Pair up the students and provide them with the engineering drawing, 3D-Printed Bar Set, masking tape, rubber bands and the small object(s) they will be picking up. Provide them with time to build each of the grippers and see how they work, testing each one's ability to pick up the various small objects. Have them share their thoughts with another group of students.

CLASSROOM IMPLEMENTATION

Questions to prompt thinking while students watch the Picking Up Pixels video:

- What do you observe or see in this professional model?
- What do you think about what you observed?
- What do your observations tell you about gripper ideas?
- What questions do you have about these gripper ideas?
- What questions would you ask the video host?

Types of Grippers

- Stationary Grippers: One finger is stationary, and the other moves.
- Geared Angular Grippers: Fingers pivot around a central point.
- Radial Grippers: Fingers move radially outward or inward.

If students feel ready, they can design their own gripper using the materials provided.

UNDERSTANDING A DECISION MATRIX (10 MINUTES)

Use the student slides to explain what a decision matrix is and how to set one up. The slides are set up to show an analysis of picking up a pencil using each of the three gripper types, modeling how to determine criteria, assigning weights, multiplying scores, summing them, and using this data to make a decision.

CREATING A DECISION MATRIX (10 MINUTES)

Pass out the decision matrix templates to each group. Have the groups of students trade the grippers they have built with another group. Using the decision matrix, one object, and the other group's grippers, have them determine which of the three gripper designs is the best. Have them document why they weighted and assigned values the way they did. Have them share back with the group they swapped grippers with.

CLASSROOM IMPLEMENTATION

Questions to guide student thinking:

- Do you feel like your decision matrix was helpful?
- What would happen if the weights of the criteria were different?
- How could the designs be improved further based on the criteria?

Evaluate and Reflect

EVALUATE (5 MINUTES)

Options to evaluate students' learning can include:

- Knowledge check questions found in the LMS course
- Presentation of their decision matrix
- Response to the following questions
    - Do you feel like your decision matrix was helpful?
    - What would happen if the weights of the criteria were different?
    - How could the designs be improved further based on the criteria?

REFLECT

Utilize the Fist to Five method, or another method of your choosing, for a quick way to gauge your student's confidence with this step of the engineering design process.

Lesson 6 | Prototyping and Iteration

Introduction

OVERVIEW

In this lesson, you will delve into the iterative process of rapid prototyping, emphasizing the importance of experimentation and learning from failures.

LEARNING OUTCOMES

In this lesson, learners will:

- Learn how to quickly create and test simple prototypes.
- Understand the value of failing early and often to improve designs.
- Share prototypes, get feedback, and use it to improve designs.

Prepare

MATERIALS

Each team will need:

- 3D Printed Bar Set (one set per two students)
- Paper
- Pens/pencils/markers
- Rubber bands
- Masking tape
- Small objects to pick up (soda can, cube, ball, etc.)
- Drinking straws (non-bendy)
- Plastic spoons
- Cardboard
- Popsicle sticks
- Cardstock
- String
- Hot glue

DOWNLOADABLE RESOURCES

- Student Facing Slide Deck
- Rapid Prototyping Video
- See-Think-Wonder Template

SETUP

- Have the 3D Printed Bar Set printed and divided out into sets *(one set per two students)*
- Print or digitally share the downloadable resources with the students.
- Have the student-facing slide deck ready to present.
- Make sure you have a method for students to document and share their gripper prototypes and iterations.

Learn and Apply

PROTOTYPING PROCESSES (10 MINUTES)

Pass out the See-Think-Wonder template, watch the Rapid Prototyping video with students, and have them fill it out. Using the slide deck, review the prototyping process with the students, including the key concepts and tips.

PROTOTYPING CHALLENGE (30 MINUTES)

Using the same pairs as the previous lesson and the materials available, students should test each of their designs and iterate on them to create the best gripper possible. Encourage them to document their ideas and each of their tests. Have groups share and get feedback to improve.

CLASSROOM IMPLEMENTATION

Questions to guide student thinking:

- How do you add complexity that improves a design rather than just making it complex for the sake of complexity?
- What materials do you not have that would be beneficial?
- What would the application of your gripper version be for a *FIRST* robot?

Evaluate and Reflect

EVALUATE (5 MINUTES)

Options to evaluate students' learning can include:

- Knowledge check questions found in the LMS course
- Presentation of their various gripper prototypes and iterations
- Response to the following questions
    - What were the most challenging aspects?
    - How did embracing failures help improve your designs?
    - What will you do differently in future prototyping activities based on what you learned today?

REFLECT

Utilize the Fist to Five method, or another method of your choosing, for a quick way to gauge your student's confidence with this step of the engineering design process.

Lesson 7 | Culminating Challenge

Introduction

OVERVIEW

In this lesson, you will utilize the engineering design process to solve a challenge utilizing 3D-printed parts.

LEARNING OUTCOMES

In this lesson, learners will:

- Identify and share your most vital skills in the engineering design process.
- Work with a group to use everyone's skills to solve a design challenge.
- Follow the engineering design process to create and present a solution to a problem.

Prepare

MATERIALS

Each team will need:

- XRP Robot (one robot per two students)
- AA batteries (four batteries per XRP Robot)
- Computer with access to the internet
- 3D Printed Bar Set (one set per two students)
- Paper
- Pens/pencils/markers
- Rubber bands
- Masking tape
- Drinking straws
- Plastic spoons
- Cardboard
- Popsicle sticks
- Cardstock
- String
- Hot glue
- Ping-pong balls
- Plastic cups

DOWNLOADABLE RESOURCES

- Student Facing Slide Deck
- Engineering Design Process Graphic
- Decision Matrix Template

SETUP

- Print or digitally share the downloadable resources with the students.
- Have the student-facing slide deck ready to present.
- Make sure you have a method for students to document and share their game concepts and/or XRP robot designs.

Learn and Apply

ENGINEERING DESIGN PROCESS REVIEW (10 MINUTES)

Take some time to review the engineering design process with students and relate it back to the previous lessons in this module. Ask the students to reflect on what areas they are most and least confident in with the engineering design process.

CULMINATING CHALLENGE (40 MINUTES)

Working together as a whole group, you will guide students in the development of a single game that they will then design a robot to play. Begin by reviewing the criteria for the game. Then let the students work in small groups to brainstorm ideas, using one of the previous strategies like mind mapping, for how the field would be set up and how you could score points using the robot, plastic cups, and ping-pong balls. Sketching is encouraged. Each small group can share their ideas with the whole group and utilize a decision matrix to determine which elements to keep in the final game.

Have the students prototype the field, assist with writing the game manual, and even them act out the game as human robots to make sure all the elements are thought through. Continue to iterate the design until the game has been finalized by the whole group.

CLASSROOM IMPLEMENTATION

Game Design Criteria:

- You should design a game using a set of small balls and plastic cups.
- It should involve two robots (alliances) versus two robots (alliances).
- It should have a real-world problem or theme to solve.
- It may involve autonomous and teleoperated control or just autonomous.
- Matches should be 2 minutes and 30 seconds.
- There should be a game manual that can be referenced with rules for the game.

Make sure to consider things like dimensions of the field. It might be best for you to create the game manual in a digital document and ask questions to guide students while recording their responses.

ROBOT DESIGN

Once they have a final game design, it is time for students to work in pairs to design an attachment for their XRP Robot to play the game following the engineering design process steps.

Evaluate and Reflect

EVALUATE (5 MINUTES)

Options to evaluate students' learning can include:

- Knowledge check questions found in the LMS course
- Presentation of their game ideas and/or XRP Robot designs
- Response to the following questions
  - What are your next steps after these initial iterations to meet your design criteria?

REFLECT

Utilize the Fist to Five method, or another method of your choosing, for a quick way to gauge your student's confidence with this step of the engineering design process.

Lesson 8 | Present and Get Feedback

Introduction

OVERVIEW

In this lesson, you will present your robot design and learn about feedback techniques.

LEARNING OUTCOMES

In this lesson, learners will:

- Identify and discuss what has been working well and what challenges a team has faced, reflecting on teamwork and progress.
- Apply effective presentation techniques to clearly communicate a team's design process and solutions, engaging the audience and using visuals.
- Participate in giving and receiving constructive feedback through the ladder of feedback and reflect on growth and areas for improvement in the engineering design process.

Prepare

MATERIALS

Each team will need:

- XRP Robot (one robot per two students)
- Computer with access to the internet
- Paper
- Pens/pencils/markers
- Tape
- Scissors
- Poster board

DOWNLOADABLE RESOURCES

- Student Facing Slide Deck
- Engineering Drawing Video
- Tips to Improve Your Presenting Skills Video

SETUP

- Print or digitally share the downloadable resources with the students.
- Have the student-facing slide deck ready to present.
- Make sure you have a method for students to create and present their final presentation with the class.

Learn and Apply

EFFECTIVE PRESENTATIONS (10 MINUTES)

No matter what type of project or what career field you are in, being able to effectively present content to an audience is an important skill. Watch the “Tips to Improve Your Presenting Skills” video and review the components of an effective presentation.

ENGINEERING DRAWINGS (5 MINUTES)

Review the elements of an engineering drawing using the “Engineering Drawings” video in the slide deck. Have students look at their previously made drawings (brainstorming for the design challenge in Lesson 1, primitive sketches in Lesson 3, gripper designs in Lesson 5, iterations in Lesson 6, game designs in Lesson 7) and see if they included these elements. Have students add them in if not already included.

BUILD YOUR PRESENTATION (30 MINUTES: PREPARE) (30 MINUTES: PRESENT)

Students will get back into their groups with their XRP Robots. Using a method of their choosing, they will develop a presentation that will later be given to the class. The presentation must include the following:

- Students’ understanding of the engineering design process and how they used it in this course
- An overview of the game (designed in Lesson 7) that the robot will play
- Their robot’s final design and former iterations, including engineering drawings

CLASSROOM IMPLEMENTATION

Determine what methods you will allow students to choose from when making their presentation. Options include poster boards, engineering notebooks, PowerPoint or other digital presentation software, a video, etc.

Evaluate and Reflect

EVALUATE (5 MINUTES)

Options to evaluate students’ learning can include:

- Knowledge check questions found in the LMS course
- Presentation of the required elements

REFLECT

Utilize the Fist to Five method, or another method of your choosing, for a quick way to gauge your student’s confidence with this step of the engineering design process.